



An in vitro comparison of mixed-reality navigation to traditional freehand and patient-specific instrumentation techniques for glenoid guide pin insertion during shoulder arthroplasty

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Background: Accurate insertion of the glenoid guide pin in shoulder arthroplasty (RSA) is important for obtaining optimized glenoid component position and orientation. The objective of this study was to evaluate and compare the accuracy of 3 glenoid guide pin insertion techniques: (1) traditional software planning using freehand guide pin insertion (freehand), (2) guide pin insertion utilizing patient-specific instrumentation (PSI), and (3) using a mixed-reality navigation (MR-NAV) system.

Methods: Twenty (20) computed tomography (CT) scans were obtained from patients exhibiting glenoid erosion patterns according to the Walch and Favard classifications. Cases were planned using validated 3-dimensional (3D) preoperative planning software. The CT data were then used to 3D print triplicate plastic models of each glenoid to evaluate the 3 guide pin insertion techniques. The first technique employed traditional software planning with freehand guide pin insertion. The second method used preoperatively planned PSI guides, whereas the third used an MR-NAV system, which provided real-time holographic guidance during guide pin insertion. Once all guide pins had been inserted into the models, an independent optical tracking system and custom digitization device was used to quantify the position and orientation of each guide pin relative to the glenoid. The outcomes for this study included the absolute mean error in guide pin inclination, version, and entry point relative to the preoperative plan. The absolute Total Global Error was also assessed, which was defined as the sum of the absolute guide pin orientation and position error relative to the preoperative plan.

Results: No statistically significant differences between MR-NAV and PSI were found for the inclination error ($2^\circ \pm 1^\circ$ vs. $2^\circ \pm 1^\circ$; $P = .056$), version error ($1^\circ \pm 1^\circ$ vs. $1^\circ \pm 1^\circ$; $P > .99$), and Total Global Error (5 ± 1 [mm + deg] vs. 5 ± 1 [mm + deg], $P > .99$), respectively. The freehand technique produced significantly greater error than MR-NAV and PSI for inclination ($5^\circ \pm 3^\circ$, $P \leq .017$), version ($4^\circ \pm 3^\circ$, $P \leq .032$), and Total Global Error (8 ± 3 [mm + deg], $P < .001$). No statistically significant differences in the entry point error were observed between all guide pin insertion methods ($P \geq .058$).

Discussion: These results demonstrate that the precision and accuracy of MR-NAV is comparable to PSI and superior to a freehand technique for glenoid guide pin insertion in vitro. Further study is needed to compare the accuracy of these techniques intraoperatively, in addition to assessing a potential learning curve between surgeons of varying experience with the MR-NAV system.

Ethics approval was obtained from Western University Health Science Research Ethics Board for the project ID 116609.

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Keywords: Mixed reality navigation; traditional software planning; patient specific instrumentation guides; shoulder arthroplasty; glenoid guide pin insertion; inclination; version

Glenoid component position and orientation are important for optimized implant function and longevity in both total (TSA) and reverse (RSA) shoulder arthroplasty. Malposition of the glenoid component in version and inclination in TSA can produce eccentric joint loading and subsequent aseptic loosening.^{11,17,19,26,35,41,45,51,53} Accurate positioning of the glenoid component in RSA is equally important. Superior inclination of the glenoid baseplate has been shown to increase the risk of aseptic loosening, scapular notching, and instability.^{4,10,15,27,31-33,42}

The first critical step in defining the position and orientation of the glenoid component during shoulder arthroplasty is the insertion of the glenoid guide pin. Several different techniques for improving the accuracy of glenoid guide pin insertion have been investigated, including the use of 3-dimensional (3D) planning software. These software programs allow the user to visualize and plan the surgical procedure on 3D renderings of the patient anatomy preoperatively. Previous studies have shown this method, when used with freehand guide pin insertion (freehand), to exhibit increased accuracy compared with using standard 2-dimensional computed tomography (CT) scans, although a high degree of inaccuracy can still occur with this technique.^{21,23} Alternatives for 3D preoperative planning include the use of patient-specific instrumentation (PSI) guides and surgical navigation. PSI guides are planned preoperatively and are designed to key onto specific preselected areas of the glenoid. Once positioned correctly, these devices physically guide the surgeon to the planned guide pin position and orientation. A variety of studies have shown these guides to be highly accurate in both cadaveric^{34,48,50} and in vivo^{8,9,13} settings; however, they are associated with increased costs and lengthy production times. Surgical navigation uses a camera and tracking system to provide surgeons real-time feedback as to their guide pin position intraoperatively. Regarding shoulder arthroplasty, some groups have reported a high degree of accuracy in guide pin positioning using this technique,^{24,38,40} although a learning curve exists to significantly reduce the operating time with this procedure.⁵² An advantage of optical navigation is that the lengthy lead time associated with the manufacturing of PSI guides is not required. However, its use in the operating room can be cumbersome as these devices are often large and difficult to transport.

The use of mixed reality for surgical assistance has seen increased popularity over recent years. With regard to

shoulder arthroplasty, previous reports have detailed its use as a visualization tool during surgery, allowing the surgeon to manipulate the surgical plan intraoperatively. However, Abdic et al¹ reported that mixed reality visualization did not significantly improve glenoid guide pin positioning in an in vitro study, as compared with a traditional freehand technique. The combination of mixed reality with navigation (MR-NAV) has also been proposed. Very few studies exist assessing the efficacy of this technology for guide pin positioning in shoulder arthroplasty. Kriechling et al^{28,29} reported low deviations between experimental and planned guide pin position and orientation in both 3D-printed glenoid models and cadavers. However, no studies have yet to compare the use of MR-NAV for glenoid guide pin insertion to any currently employed techniques. It was the objective of this study, therefore, to compare MR-NAV to PSI guides and traditional software planning with freehand guide pin insertion (freehand). It was hypothesized that MR-NAV would increase the accuracy of guide pin insertion compared with the freehand technique and would exhibit similar accuracy to that of PSI.

Materials and methods

Preoperative planning and glenoid model creation

The CT scans from 20 patients (mean age: 68 ± 11 years) who were planned for either TSA ($n = 5$) or RSA ($n = 15$) were obtained for this study. The total sample size was deemed sufficient based on previous benchtop studies investigating different techniques for glenoid guide pin insertion.^{1,29,50} Inclusion criteria consisted of primary glenohumeral arthritis, cuff tear arthropathy, and massive irreparable rotator cuff tears. Proximal humeral fractures and cases of glenohumeral instability were excluded. Scans were also excluded if glenoid erosion had extended superiorly into the coracoid, as the coracoid served as a landmark during the MR-NAV registration process. All glenoid erosion patterns were characterized by the senior surgeon, who has more than 15 years of experience in shoulder arthroplasty. Using the Walch² and Favard⁴⁷ classifications for glenoid erosion, the following erosion patterns were obtained for this study: 2-A2, 1-B1, 6-B2, 2-B3, 2-C, 1-D, 2-E0, 2-E2, and 2-E3. All scans were uploaded to a validated^{5,36,46,49} preoperative planning software (Blueprint Mixed Reality, v. 4.0.2.; Stryker, Kalamazoo, MI, USA) where the scapula was automatically segmented. Each case was then planned by the senior surgeon (G.S.A.). TSA was planned for the single B1 glenoid and 4 of the B2 glenoid erosions. RSA

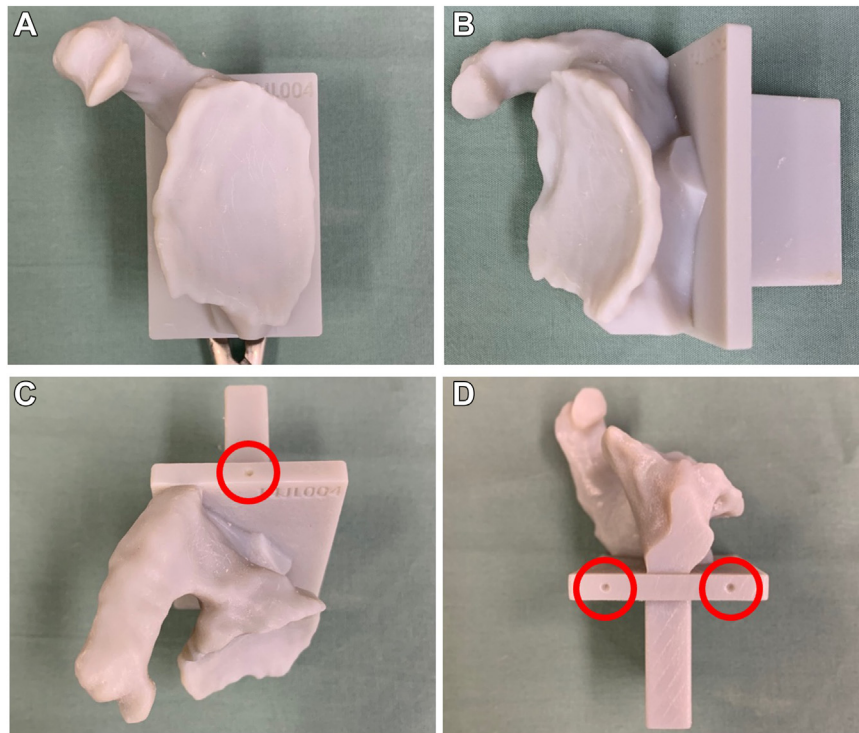


Figure 1 A three-dimensional (3D) printed model of a B2 glenoid from a lateral (Panel A), posterolateral (Panel B), superior (Panel C), and inferior (Panel D) view point. The conical indents on the superior and inferior peripheral surfaces of the base are indicated with the (○) and served to generate a glenoid coordinate system.

was planned for all other cases, including the 2 remaining cases with B2 glenoid erosions. All TSA cases were planned to reproduce inclination between 0° and 10° superior and version between 0° and 5° retroversion, whereas RSA cases were planned for neutral inclination and 0° - 10° of retroversion of the glenoid component. The planned averages for glenoid component inclination and version were $0^\circ \pm 5^\circ$ and $-4^\circ \pm 2^\circ$ for TSA cases, and $0^\circ \pm 1^\circ$ and $-6^\circ \pm 5^\circ$ for RSA cases, respectively.

Each scapula was exported as a stereolithography (STL) file into a computer-aided design software (SOLIDWORKS; Dassault Systèmes, Vélizy-Villacoublay, France) with an average triangle edge length of 0.9 mm. Within this software, the glenoid and coracoid were then sectioned from the rest of the scapular body using a plane positioned 20 mm medial to the glenoid center point and perpendicular to the transverse axis. Twenty millimeters was used as this allowed for a sufficient volume of the glenoid vault to be modeled for adequate guide pin fixation experimentally.³ This distance also contained the coracoid base attachment to the superior glenoid in all scapulae, which was needed for MR-NAV registration. Furthermore, scapular bone medial to the glenoid and coracoid is not visible during shoulder arthroplasty because of the surrounding soft tissue and therefore was not included in these glenoid models. A base was added to the medial aspect of each glenoid model to serve as a point of fixation during testing. Three conical indents were designed onto the superior and inferior faces of the rectangular base, each with a 2-mm diameter and depth. These points were used to develop a subject-specific glenoid coordinate system experimentally, which was then used in the determination of the guide pin position and orientation. Triplicate models of each glenoid

were 3D printed (Stratasys J8 series printer; Stratasys, Rehovot, Israel) (Fig. 1). To mimic cortical bone, a rigid polymer (DraftGrey; Stratasys) was printed around the periphery of the glenoid model, including the glenoid surface. A less rigid polymer (SUP706B; Stratasys) was used to fill the interior of each model and served to represent the cancellous bone within the glenoid. These models were employed in this study as this allowed for a repeated measures protocol to be employed between the 3 guide pin insertion techniques assessed. Similar models have previously been used to assess different techniques for guide pin insertion.^{1,29}

Experimental setup and procedure

Each glenoid model was fixed to a benchtop testing apparatus using a clamp. The glenoid was aligned so that the scapula transverse axis was positioned parallel to the floor. An artificial soft tissue shoulder model was placed over each glenoid model before testing. Windows were created to represent the deltopectoral interval (Fig. 2). A drill (Milwaukee Tool; Brookfield, WI, USA) was used to insert a threaded 2.5-mm Kirschner wire for all glenoid models.

All guide pin insertions were performed by a fellowship-trained shoulder surgeon (R.G.) who was blinded to all the pre-operative plans. This surgeon had previously performed more than 200 shoulder arthroplasty surgeries on a variety of pathologies and glenoid morphologies using both freehand and PSI techniques, with limited experience using MR-NAV due to the novelty of this system. However, sufficient training with the MR-NAV system in

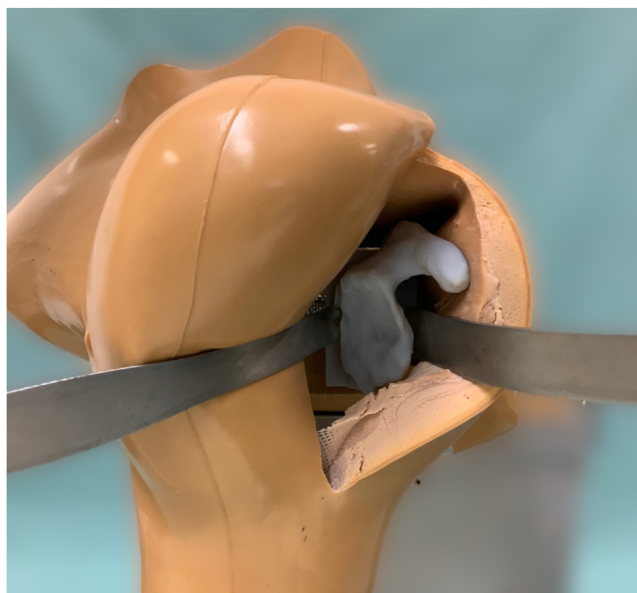


Figure 2 The experimental shoulder guide pin insertion setup. The glenoid model is affixed to the testing base using a c-clamp (not shown). An artificial soft tissue model is used to cover the glenoid model, with glenoid exposure obtained using a simulated deltopectoral approach.

this study was provided before testing using glenoid models not used in this study.

Three techniques for glenoid guide pin insertion were evaluated. The first used a freehand guide pin insertion technique, which consisted of traditional software planning and guide pin insertion without any assistive devices. The surgeon was allowed to view and manipulate the preoperative plan during this process using a computer and monitor positioned behind the testing apparatus.

The second guide pin insertion technique used PSI guides (Blueprint 3D Planning Software; Stryker) coupled with a pin cannula. The PSI guide design used in this study has previously been shown to be accurate and reproducible in obtaining planned guide pin position in both cadaveric⁵⁰ and in vivo studies.¹³ Each guide consisted of 4 legs that were fitted to 3 points on the anterior glenoid rim and 1 point on the posterior glenoid rim (Fig. 3). The location of each of these points were planned preoperatively by the senior author. All guides were exported from the preoperative planning software and 3D printed from titanium to minimize deflection of the guide during guide pin insertion. During the insertion process, the surgeon was allowed to view the preoperative plan, which illustrated the placement of the PSI guide relative to the glenoid.

The final guide pin insertion technique evaluated in this study used an MR-NAV system (Blueprint Mixed Reality Guidance with Holobluemprint software; Stryker). This system used a Hololens II (Microsoft, Redmond, WA, USA) head-mounted display, which provided holographic visualization. This system also consisted of a guide handle, stylus, and coracoid clamp (Fig. 4). The coracoid clamp was first securely fixed to the tip of the coracoid. A quick response cube was rigidly attached to this clamp, which served as the reference marker throughout the navigation process. This clamp was orientated such that 3 faces of the cube were visible to

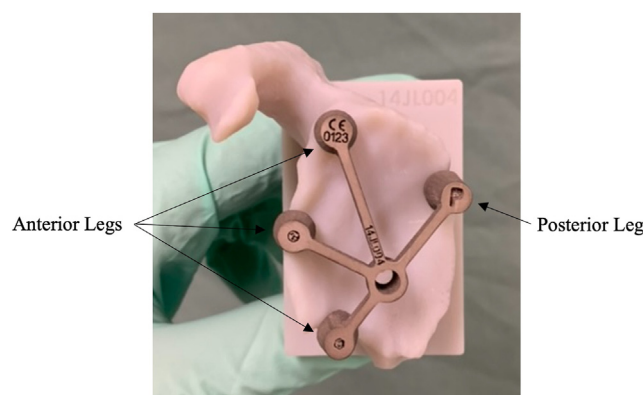


Figure 3 Metallic PSI guide fitted to glenoid model. Three legs are positioned on the anterior glenoid rim whereas 1 leg is positioned on the posterior glenoid rim. *PSI*, patient-specific instrumentation.

the camera of the head-mounted display. Registration was then performed using the stylus. Five points on the glenoid, consisting of superior, inferior, anterior, posterior, and center points, were digitized in addition to the coracoid tip. This was followed by the collection of 3 traces: the glenoid surface, coracoid base, and coracoid tip. Once all points and traces had been collected, the surgeon was able to view the registration accuracy across the various regions of the glenoid and coracoid. If the registration error was greater than 2 mm on any of the digitized surfaces, the surgeon was instructed to reperform either a specific trace or the entire registration process depending on the location of the errors. Once the surgeon was satisfied with the global registration accuracy, guide pin insertion was then performed. For the MR-NAV procedure, real-time holographic feedback was provided in 2 phases (Fig. 5). First, the anterior to posterior (AP) and superior to inferior (SI) translation of the entry point location on the glenoid was displayed to the surgeon. Once the guide pin entry point location had been selected, the surgeon was then provided feedback as to the planned guide pin inclination and version. Although the surgeon was only provided 1 attempt to insert the guide pin into the glenoid model, they were able to repeat any step of the registration or alignment process as many times as they deemed necessary for achieving stable and accurate guide pin insertion.

Once all guide pins had been inserted, an optical tracking system (Optotrak Certus; Northern Digital, Waterloo, ON, Canada) and custom digitization device were used to quantify the position and orientation of each inserted guide pin relative to the glenoid (Fig. 6, A). This digitization device was designed to allow for rotation and translation of this device along the axis of the inserted guide pin, while preventing undesired translation in a plane normal to the guide pin axis. Furthermore, this device contained 3 digitization holes on its top surface that were located a known distance away from the central guide pin axis. To quantify the axis vector of the inserted guide pin, each hole was digitized using a calibrated stylus as the device was rotated about the inserted guide pin (Fig. 6, B). During this procedure, the digitization device was positioned as close as possible to the glenoid model to minimize potential deflection of the guide pin. A 3D circle was then fit to each trace to determine a vector representing the central axis of the guide pin (circle normal) and a point on this axis (circle center). To assess the error in quantifying the guide pin axis using this technique, the radius of each best-fit 3D circle was

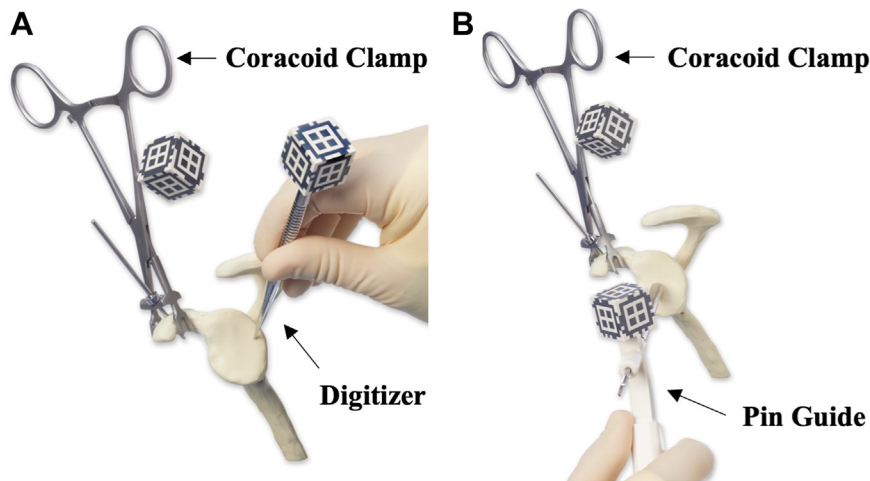


Figure 4 The handheld equipment used during the MR-NAV guide pin insertion process. Panel **A** illustrates the use of the digitizer and panel **B** illustrates the use of the handheld pin guide. *MR-NAV*, mixed-reality navigation.

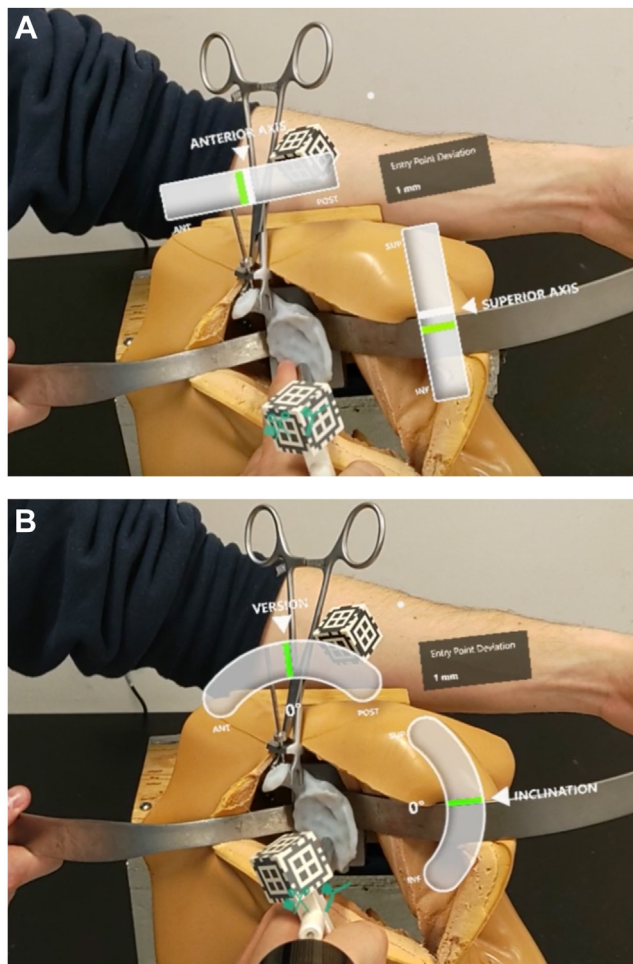


Figure 5 The real-time holographic navigation provided by mixed reality. Panel **A** demonstrates the surgeon's view of the holograms that provide real-time feedback for glenoid entry point selection. Panel **B** shows the surgeon's view for glenoid inclination and version guidance.

compared to the theoretical distance between the apex of each digitization hole and central guide pin axis. A mean absolute difference of 0.3 ± 0.4 mm was found and was deemed acceptable for accurate quantification of the guide pin position and orientation.

To quantify the relative position and orientation of the guide pin axis relative to the glenoid model, the location of the indents on the periphery of each glenoid model's base was additionally digitized to establish an experimental glenoid coordinate system for each model. This coordinate system was used to transform the guide pin central axis vector into the preoperative planning software's frame of reference. This allowed for the direct comparison of the experimental guide pin position and orientation to that planned preoperatively.

Outcome variables and statistical analysis

The primary outcomes for this study were the absolute error in guide pin inclination, version, and entry point. Guide pin insertion and version were determined using methods as described in Boileau et al.⁵ The entry point of all experimental guide pins was determined using a ray-projection algorithm using the experimental guide pin central axis and a point located on this axis. To determine the entry point of the pin on the glenoid face, a ray was projected along the direction of the guide pin axis vector from a point located on the guide pin. The entry point was calculated as the point of interception between the ray and the glenoid STL model. An additional variable, the Total Global Error, was also calculated. This descriptive metric was defined as the sum of the entry point and total orientation errors for each experimental guide pin. Although the clinical impact of this variable has not yet been validated, this descriptive metric was reported not to provide clinically relevant insight but to express in a general manner the total error associated with guide pin insertion. Statistical significance was assessed for each outcome variable using a 1-way repeated measures analysis of variance (SPSS; IBM, Armonk, NY, USA). The 3 guide pin insertion techniques served as the independent variables. A Bonferroni correction was applied to

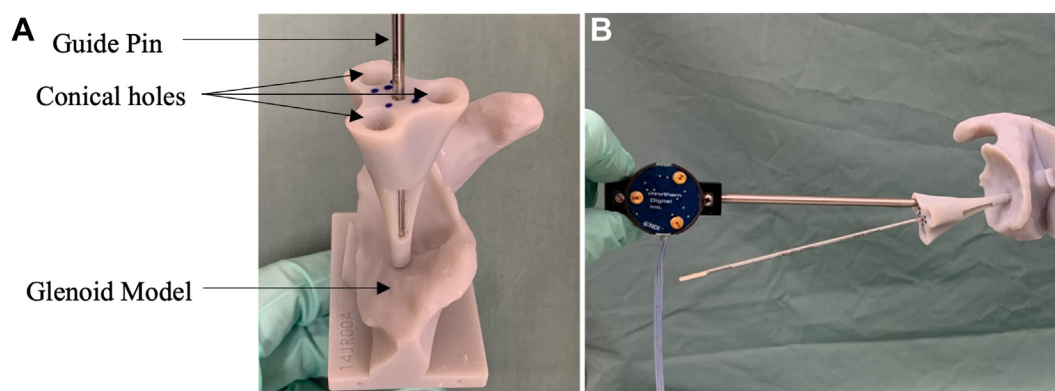


Figure 6 (A) The custom digitization device was positioned on the inserted glenoid guide pin. This device consists of 3 conical indents on its top surface located along a known diameter relative to the central axis of the guide pin. (B) A stylus with a rigidly attached optical tracking marker is used to digitize the position of each of the conical indents.

account for multiple comparisons, and the significance was set at a threshold of $P \leq .05$.

Results

The accuracy of each of the 3 guide pin insertion techniques for all outcome variables are illustrated in Figure 7, whereas the maximum errors associated with each technique are presented in Table I. The mean absolute inclination errors using freehand, PSI, and MR-NAV guide pin insertion techniques were $5^\circ \pm 3^\circ$, $2^\circ \pm 1^\circ$, and $2^\circ \pm 1^\circ$ respectively. The mean absolute inclination error using the freehand technique was significantly greater than that using PSI ($P = .017$) and MR-NAV ($P = .001$). Similar significance was observed with the mean version error for each method. The mean version errors using freehand, PSI, and MR-NAV techniques were $4^\circ \pm 3^\circ$, $1^\circ \pm 1^\circ$, and $1^\circ \pm 1^\circ$, respectively. Statistically significant differences were again observed between the freehand technique and both PSI ($P = .031$) and MR-NAV ($P = .035$). Regarding the guide pin entry point, similar error was observed between all techniques with an absolute error of 2 ± 1 mm, and no statistically significant differences were detected ($P \geq .058$).

For the Total Global Error outcome, no statistically significant differences were observed between MR-NAV and PSI techniques, with an absolute error of 5 ± 1 [mm + deg] detected for both ($P > .99$). The greatest Total Global Error occurred with the freehand technique, with an error of 8 ± 3 [mm + deg], which was significantly greater than both MR-NAV and PSI ($P < .001$).

Discussion

The results of this study demonstrate that PSI and MR-NAV are both equally accurate techniques for obtaining planned

glenoid guide pin trajectory in shoulder arthroplasty. Both assisted techniques exhibited significantly less error in both guide pin inclination and version error compared with the freehand technique. Perhaps more importantly, both PSI and MR-NAV techniques exhibited substantially less maximum error in both inclination and version compared with the freehand method (Table I). This is likely due to the real-time guidance each technique provides the surgeon. The MR-NAV system provides the user real-time feedback of the guide pin position and orientation based on the location of the guide handle tracker relative to the coracoid tracker, and the information obtained from the registration process. If the registration process is performed correctly and with minimal error, then the final accuracy of the guide pin insertion is dependent on the surgeon's ability to maintain accuracy while inserting the guide pin into bone. Conversely, PSI guides rely solely on physical guidance. The PSI guide must be positioned correctly on the face of the glenoid intraoperatively, as the guide pin is physically drilled through the PSI guide and into the bone. Guide pin insertion with the freehand method, however, was observed to exhibit the greatest error in pin inclination and version relative to the preoperative plan. This is likely due to the surgeon having to visually estimate the guide pin orientation relative to the 3D glenoid after viewing the 3D preoperative plan without any physical guidance or feedback.

Although PSI and MR-NAV exhibited increased accuracy compared to freehand insertion for guide pin inclination and version, similar error was observed across all 3 methods for the guide pin entry point. A potential explanation as to why the freehand method did not exhibit greater entry point error compared to PSI and MR-NAV techniques may stem from the glenoid erosion patterns used in this study. The complexity of the glenoid erosions evaluated may have provided the surgeon with several unique bony landmarks on the glenoid face, which may have aided the surgeon in determining the planned guide pin entry point relative to the preoperative plan.

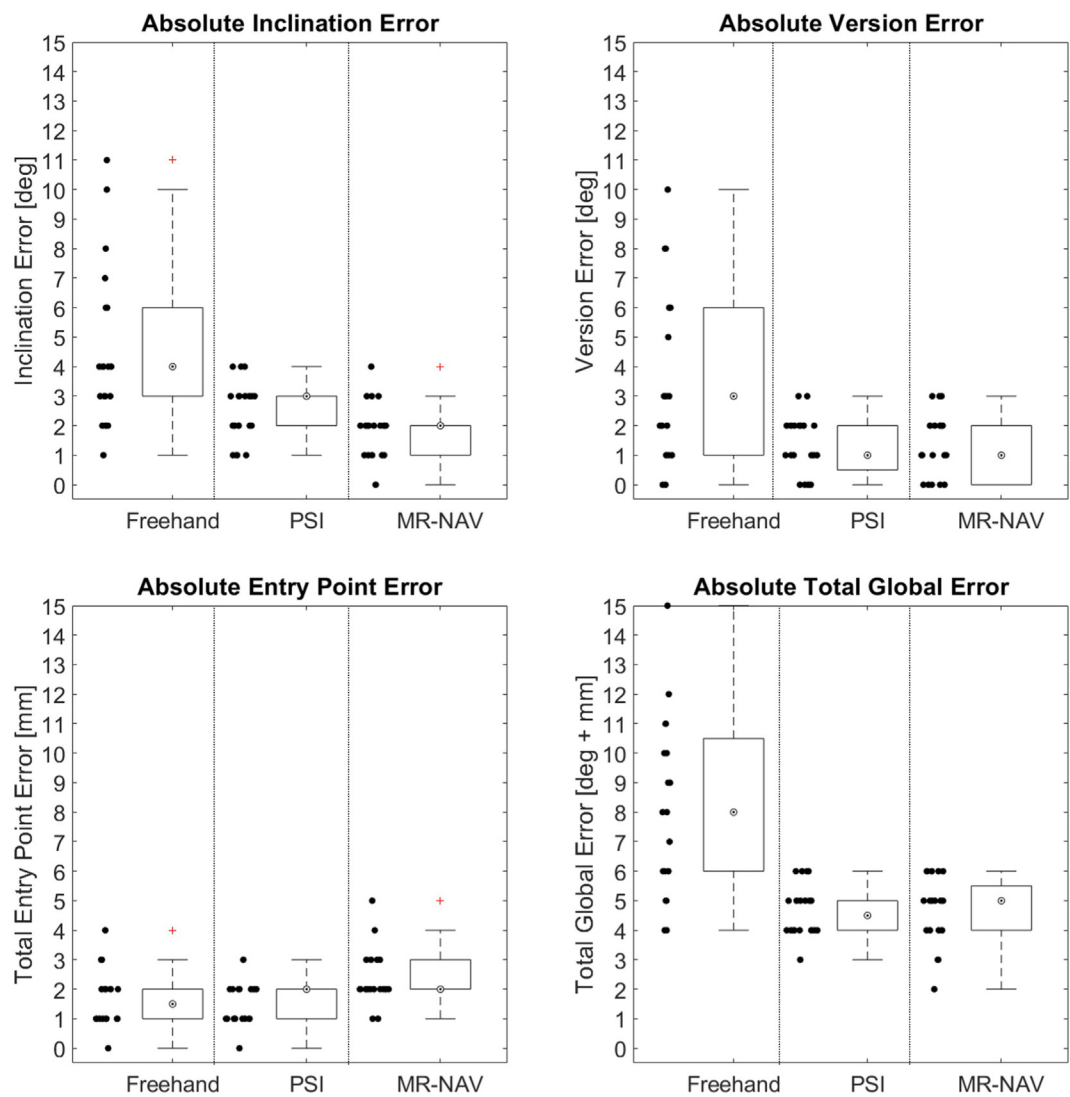


Figure 7 Box and whisker plots illustrate the absolute error in guide pin inclination (*top left*), version (*top right*), entry point (*bottom left*), and Total Global Error (*bottom right*) in Freehand, PSI, and MR-NAV techniques. *PSI*, patient-specific instrumentation; *MR-NAV*, mixed-reality navigation; *deg*, degrees.

Table I Absolute maximum errors observed for all guide pin insertion techniques and outcome measures				
	Inclination, degrees	Version, degrees	Entry point, degrees	Total Global Error, degrees + mm
Freehand	11	10	4	15
PSI	5	3	3	6
MR-NAV	4	4	5	6

PSI, patient-specific instrumentation; *MR-NAV*, mixed-reality navigation.

To our knowledge, this is one of the first studies that compared the use of MR-NAV to standard guide pin insertion techniques for shoulder arthroplasty. Kriechling

et al have evaluated an MR-NAV system in both 3D-printed scapulae²⁹ and cadavers.²⁸ Their group used an in-house registration system, which followed a similar navigation process as used in our study. Registration was first performed by collecting points on the glenoid and coracoid followed by traces collected on the glenoid surface and coracoid base. A standard reference custom-made tracker was then attached to a drill sleeve to assist in the insertion of the guide pin, allowing real-time visual guidance to be obtained. Mean entry point error and 3D orientation errors of 2 mm and 3° were reported for the 3D-printed scapulae study, and 4 mm and 4° for the cadaveric study, respectively. These studies, however, did not report on inclination and version error. Furthermore, all testing was completed using nonarthritic glenoids, which may have influenced the results obtained in their study. Finally, they did not compare

their holographic navigation to standardized techniques such as PSI, optical navigation, or freehand.

Abdic et al¹ evaluated glenoid guide pin accuracy using mixed reality visualization compared to freehand insertion in a dry lab benchtop study. The mixed reality visualization system they used only allowed viewing of the preoperative plan, with no navigation component to assist in guide pin insertion. The authors reported that mixed reality visualization provided no statistically significant benefit in either procedure time or guide pin accuracy. When compared to the results of this present study, these findings emphasize the importance of the navigation aspect in a mixed reality system to provide quantitative benefits in guide pin insertion accuracy.

The accuracy of guide pin insertion using the freehand technique in this study is similar to previous literature. Abdic et al¹ in their dry lab study reported guide pin inclination, version, and entry point error using freehand guide pin insertion to be greater than or equal to 6°, 5°, and 2 mm, respectively, for all surgical levels evaluated. Similarly, Iannotti et al²¹ in another dry lab study reported mean inclination, version, and entry point errors using 3D preoperative planning of 9°, 7°, and 2 mm. The use of 3D preoperative planning has been shown to increase the accuracy of guide pin inclination and version by as much as 5° compared with 2-dimensional CT scans.²¹ However, our results combined with those from previous studies support that 3D preoperative planning alone is not sufficient to achieve highly accurate and reliable insertion of the glenoid guide pin.^{21,23}

Increased guide pin insertion accuracy using PSI guides has also previously been reported. Cadaveric studies evaluating PSI guides for glenoid guide pin insertion have reported inclination and version errors of less than 3°, and entry point errors of less than 2 mm.^{34,50} Several in vivo studies have also reported increased accuracy in guide pin insertion when using PSI guides.^{8,9,16} However, a clinical study from Lau et al³⁰ reported poor accuracy using PSI guides in patients undergoing both TSA and RSA. In their study, the authors found mean absolute deviations in inclination and version of 4° and 11° for TSA, and 9° and 10° for RSA. The patients in the study from Lau et al had severe posterior glenoid erosions (mean retroversion of 22°), which may have increased the difficulty of correcting version to neutral during the procedures. Additionally, PSI guides are associated with lengthy production times. This may favor the use of MR-NAV as this technology can be incorporated into the preoperative planning workflow, thus making it more accessible to surgeons with less lead time required.

Accurate insertion of the guide pin is an important first step to ensuring optimized glenoid component placement in both TSA and RSA. This is especially true in cases with severe glenoid erosion, which increases the difficulty in replicating the preoperative plan using a freehand technique.²² Deviation in guide pin position from the

preoperative plan can result in glenoid component malposition and lead to detrimental consequences. In TSA, glenoid component malposition in version can cause humeral head subluxation and eccentric joint loading, which can lead to subsequent aseptic loosening through a rocking effect of the glenoid component.^{11,17,19,35,41,45} Similarly, when the glenoid component is positioned in superior inclination, this can lead to superior translation of the humeral head in patients with a functioning rotator cuff.⁵¹ Such translation can lead to increased rotator cuff and deltoid loading during abduction and eccentric joint loading, and thus increase the risk for component loosening and superior rotator cuff tears.^{26,51,53} Griffin et al¹⁴ additionally reported improved clinical outcomes when glenoid inclination is corrected to within 2° of preoperative measurements.

In RSA, it is recommended that superior inclination of the base plate be avoided as this has been shown to produce eccentric joint loading and lead to subsequent component loosening.^{4,15,27,31} Furthermore, superior inclination and superior placement of the baseplate is associated with scapular notching, which can lead to polyethylene wear and increase the risk for component loosening.^{10,32,33,42} Use of an anterosuperior approach in RSA³⁷ and superior glenoid erosion⁷ are risk factors for the placement of superiorly inclined glenoid components and may benefit from additional surgical assistance for guide pin insertion. Boileau et al⁶ formulated the RSA angle that served to provide an alternate assessment of inclination of the inferior glenoid where the base plate is often positioned. In their imaging study, the authors recommended that the glenoid baseplate should be positioned with an RSA angle close to 0° as this should increase the efficiency of the remaining cuff and reduce eccentric loading on the glenosphere. Less is known regarding optimal placement of the glenoid component in version, with previous work showing significant variance to exist between surgeons in correcting version in RSA.⁴³ However, some computational studies have found components placed in neutral or slight retroversion to optimize range of motion and improve stability.^{12,25,44}

Early basic science evidence,^{28,29} including the results from this study, suggest MR-NAV to be accurate in obtaining planned guide pin position. As such, further studies are needed to evaluate the accuracy of this technique intraoperatively, as it is currently unclear whether the increased accuracy observed in these biomechanical studies will translate to a clinical setting. Furthermore, the potential effects of this increased accuracy on implant longevity and clinical outcomes must also be investigated, as the clinical benefits associated with PSI and computer navigation have recently been called into question.^{18,20,39}

Some limitations were present in this study. Testing was conducted in a laboratory setting with plastic 3D glenoid models and likely did not replicate the difficulty in obtaining adequate glenoid exposure as experienced in the operating room due to potential soft tissue impingement.

Furthermore, these 3D-printed models perfectly represented the osseous anatomy of the glenoid as obtained on the CT scan. In the operating room, any remaining glenoid cartilage or soft tissue would need to be removed prior to the commencement of guide pin insertion to minimize the risk of impacting the seating of the PSI guides, altering the appearance of the glenoid, and influencing the trace of the glenoid during registration using MR-NAV. Additionally, this study did not consider the potential influence further surgical steps could have on the final position and orientation of the glenoid component, such as glenoid reaming, central peg drilling, screw insertion, and the positioning of any potential augment.⁴⁰ This study was also limited in sample size to perform a thorough comparison between mild and severe erosion types. It is possible that both PSI and MR-NAV techniques may provide greater benefit to patients with more severe glenoid bone loss. As such, further investigation is needed. Lastly, all guide pin insertions were performed by a single fellowship-trained surgeon throughout this study. Previous work has shown surgeon experience not to significantly impact planned guide pin position or orientation accuracy for both freehand and assisted techniques,¹ whereas the utilization of a single surgeon has also previously been used in similar biomechanical studies.^{29,50} However, further investigation is needed to assess whether surgeon experience influences guide pin insertion accuracy, and possibly the learning curve, when using an MR-NAV system.

Conclusion

MR-NAV and PSI are accurate techniques compared with 3D preoperative planning with freehand guide pin insertion for obtaining planned guide pin position and orientation during shoulder arthroplasty. Further study is needed to assess the relative accuracy of each of these methods in a clinical setting and the potential differences in implant longevity and complication rate following these techniques.

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